Supplementary material

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Supplement to "Annotating single-cell data on a laptop". Figures and tables referenced from the main text as Supplementary Figure S1–S4 and Supplementary Table S1–S3.

1 Supplementary figures

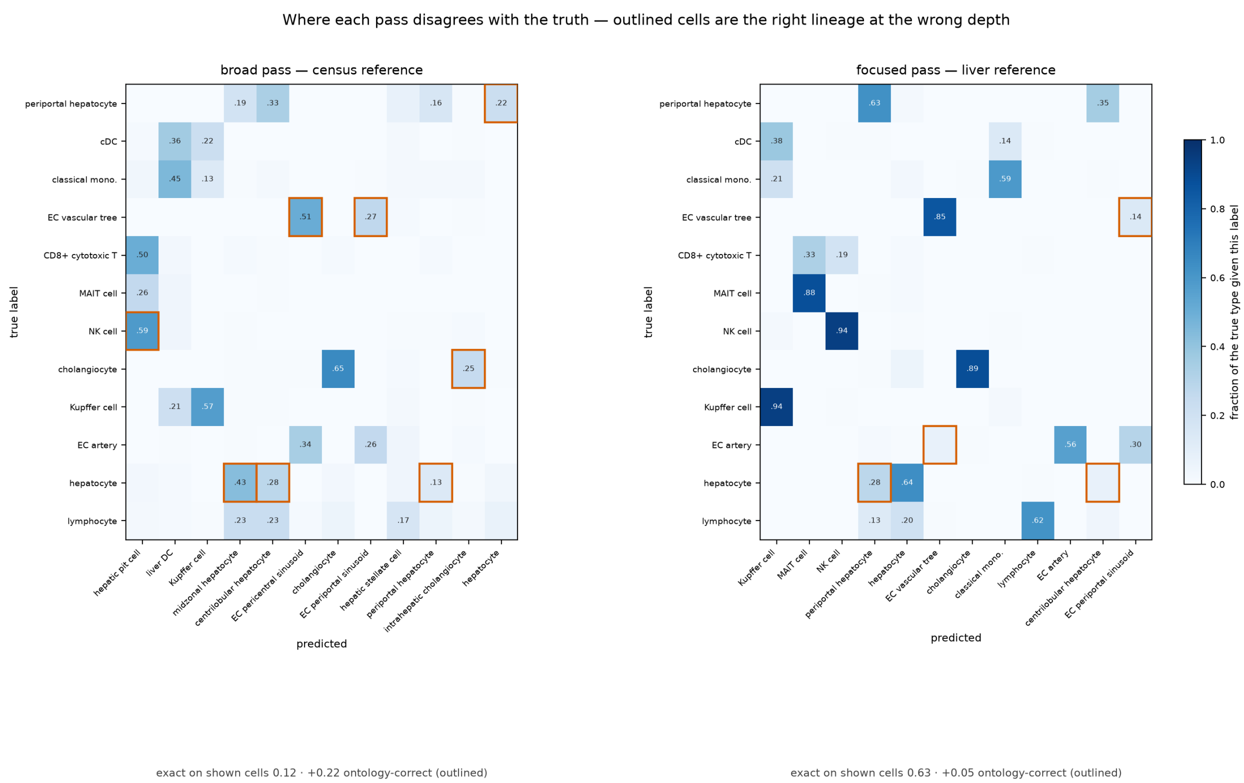

Figure S1. Confusion matrices for the broad and focused passes on the withheld cross-study liver query. Each cell is the fraction of a true type receiving that label; outlined cells are off-diagonal calls that are an ancestor or descendant of the truth in the Cell Ontology, i.e. error under exact match and credit under ontology-aware concordance. The scores below each panel are means over the twelve truth types drawn, not over the whole query, so they run lower than the query-wide figures quoted in the main text. The broad pass scores 0.12 exact and recovers +0.22 through the ontology — its mistakes are the right lineage at the wrong depth (hepatocyte → midzonal or centrilobular hepatocyte; NK cell → hepatic pit cell, the Cell Ontology term for a liver NK cell). The focused pass scores 0.63 exact, +0.05: it is already right most of the time, so the ontology has little left to recover.

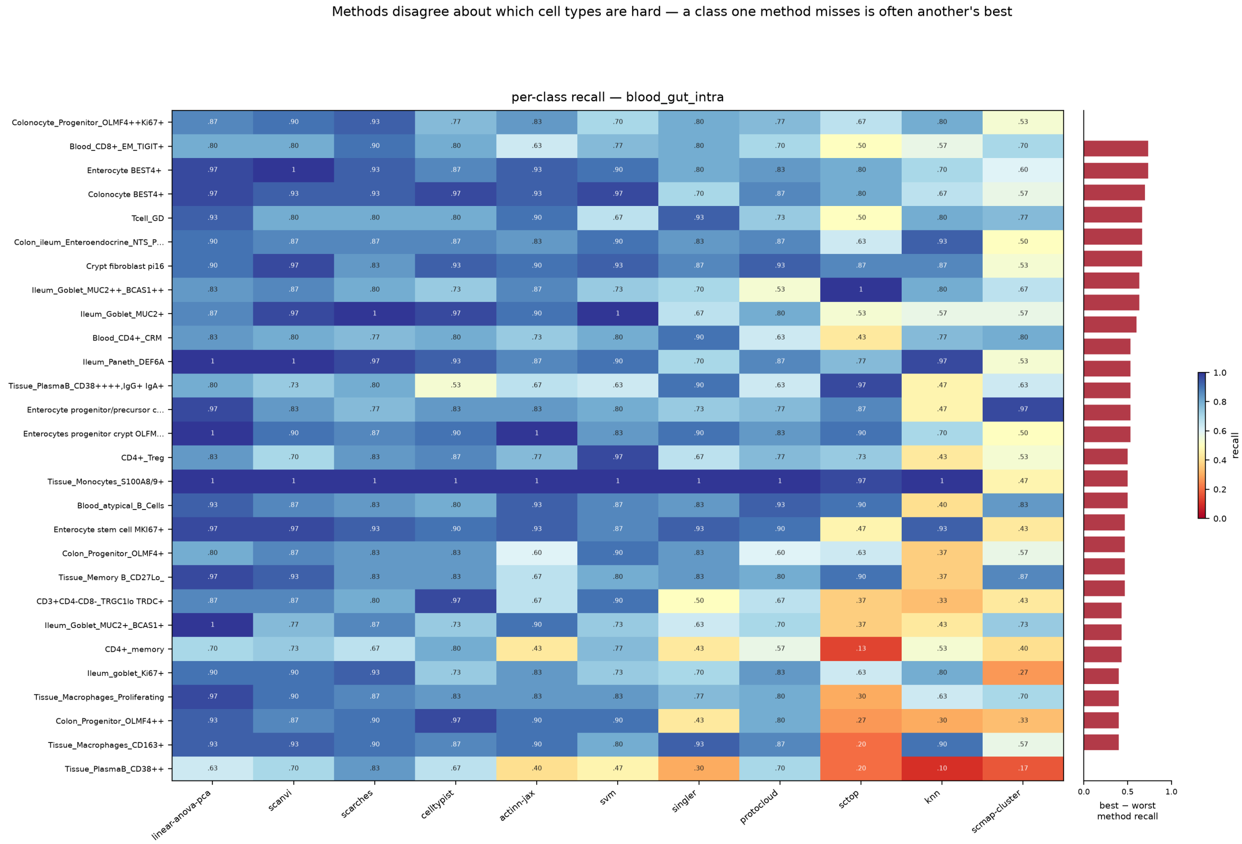

Figure S2. Per-class recall for eleven methods on the 86-type blood+gut split, ordered by how much the methods disagree (best minus worst recall, right-hand strip). The best and worst method differ by a median of 0.30 recall per class and up to 0.73; mean pairwise Spearman between methods' per-class recall is 0.58. Class size is not the explanation — the test split is capped per label at 14–30 cells for all 86 classes, and the ten smallest classes score slightly higher than the ten largest (0.865 vs 0.827).

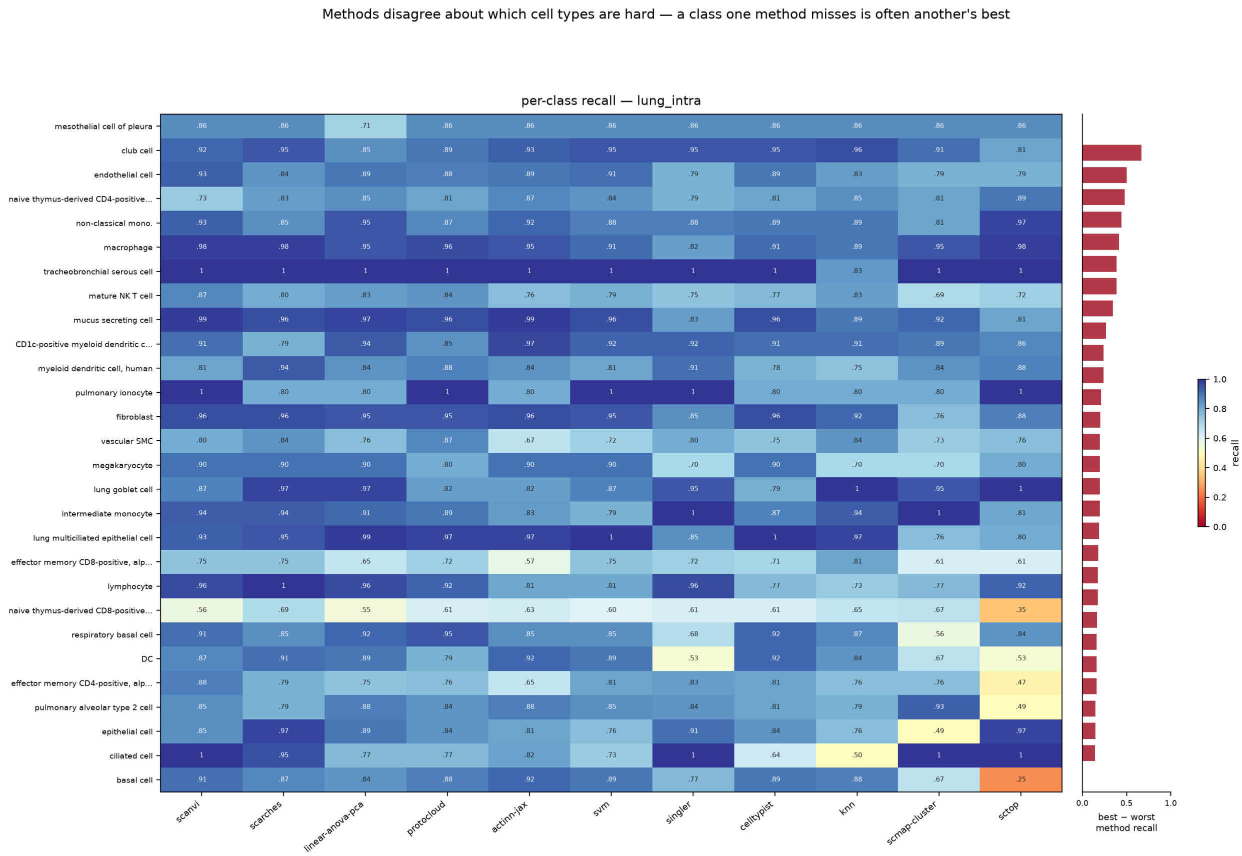

Figure S3. Per-class recall on the 46-type lung split, drawn as in Figure S2. Methods agree more here than on blood+gut: mean pairwise Spearman 0.65, median best-minus-worst 0.16.

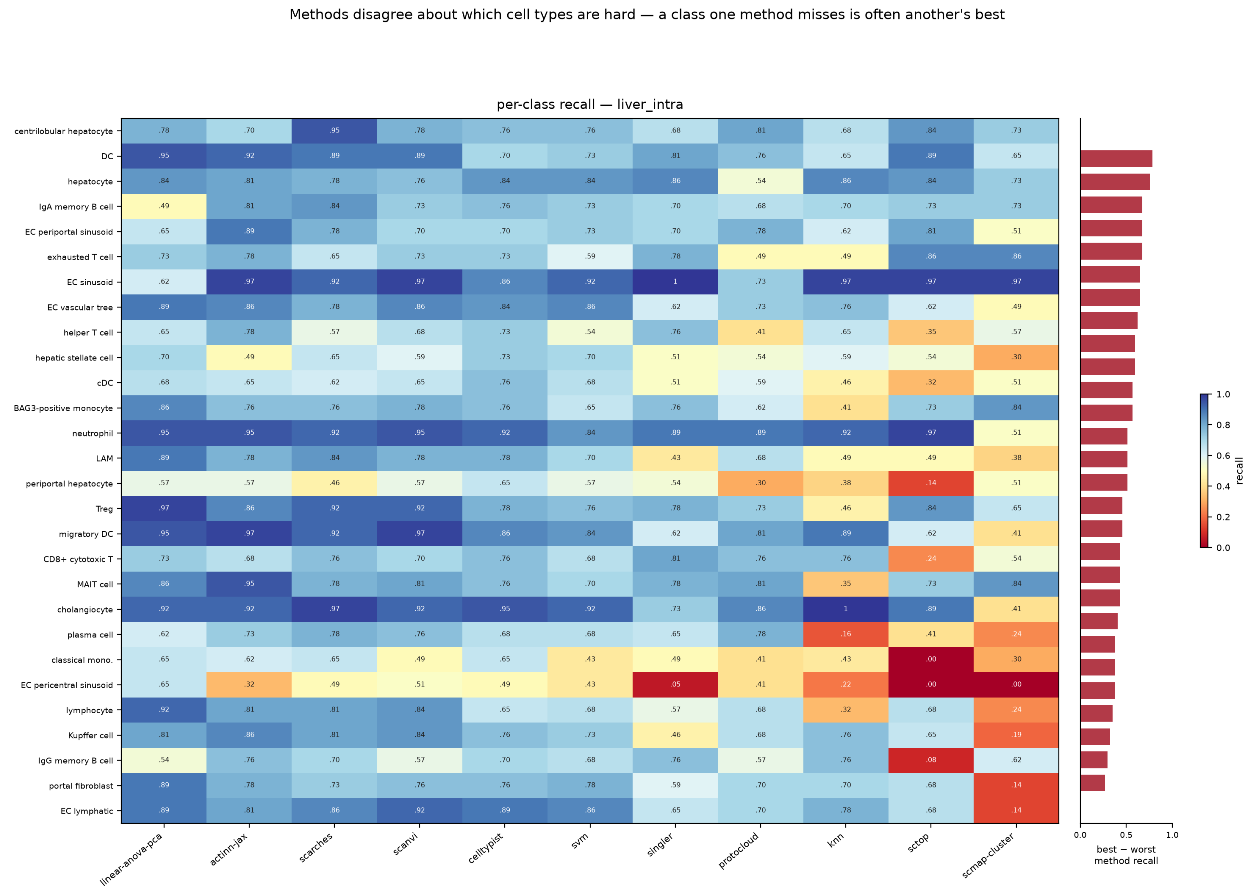

Figure S4. Per-class recall on the 36-type liver split, drawn as in Figure S2. Mean pairwise Spearman is 0.66, but the median best-minus-worst gap is 0.43 — the widest disagreement of the three splits, and a reminder that a high rank correlation between methods still leaves large per-class differences.

2 Supplementary tables

method	source	tier	engine	rejection	runtime
actinn-jax	[Ma & Pellegrini 2020]	classical	JAX MLP	confidence threshold	JAX
SVM	[Pedregosa 2011]	classical	linear SVM (SGD)	—	scikit-learn
kNN	[Pedregosa 2011]	classical	k-nearest neighbors	—	scikit-learn
CellTypist	[Domínguez Conde 2022]	linear	L2 logistic regression	prob threshold	scikit-learn
linear-anova-pca	[Pedregosa 2011] †	linear	normalize → ANOVA → PCA(220) → logreg	prob	scikit-learn

method	source	tier	engine	rejection	runtime
scTOP	[Yampolskaya 2023]	parameter-free	rank z-score class-average projection	—	NumPy
SingleR	[Aran 2019]	correlation	Spearman + fine-tuning	—	R
scmap-cluster	[Kiselev 2018]	correlation	centroid cosine	yes (unassigned)	R
scANVI	[Xu 2021]	deep	scVI semi-supervised VAE	prob	scVI
scArches	[Lotfollahi 2022]	deep	scANVI reference surgery	prob	scVI
ProtoCloud	[Guo & Ding 2026]	deep	prototype VAE + LRP attribution	ambiguity flag	PyTorch
scPRINT	[Kalfon 2025]	foundation	pretrained transformer, zero-shot	—	PyTorch
Pan-human Azimuth	[Sarkar et al. 2026]	pretrained	8-level hierarchical NN, fixed 382-leaf typology	trained `Unassigned`	TensorFlow

Table S1. The benchmarked methods: model family (*tier*), the engine each one runs, whether it can decline to call a cell (*rejection*), and the runtime stack it was run on. Bold marks the methods added to this panel here.

† `linear-anova-pca` has no method paper: it is a baseline assembled here from scikit-learn components, tuned deliberately to be the strongest simple competitor we could build.

dataset	source	split	tissue	#types	genes	notes
lung_intra	[Travaglini 2020]	within-dataset	lung (Krasnow)	46	Ensembl	300 cells/type ref
lung_cross	[Sikkema 2023] → [Travaglini 2020]	cross-dataset	lung (HLCA → Krasnow)	46	Ensembl	different lab/protocol
liver_intra	[Edgar 2026]	within-dataset	liver (HLiCA)	36	Ensembl	150 cells/type
liver_cross	[Edgar 2026]	cross-**study**	liver (HLiCA)	34	Ensembl	train 6 studies → test withheld study
blood_gut_intra	[Alegbe 2026]	within-dataset	blood + gut (IBDverse)	86	Ensembl	high cardinality; no CL ids

dataset	source	split	tissue	#types	genes	notes
pbmc	[10x Genomics 2016]	within-dataset	PBMC (pbmc3k)	8	symbols	small-n; scPRINT skips (symbols)

Table S2. The six benchmark datasets. *split* separates within-dataset holdouts from cross-dataset and cross-study transfer; *genes* records whether the matrix is keyed by Ensembl ids or gene symbols.

dataset	actinn-jax	best method (acc)
lung_intra (46 types)	0.894	ProtoCloud 0.932
lung_cross (cross-dataset)†	0.358 exact / 0.752 ontology	ProtoCloud 0.374 / **0.791 ontology**
liver_intra (36 types)	0.802	linear-anova-pca 0.804
liver_cross (cross-study)	**0.686 exact / 0.731 ontology**	actinn-jax
blood_gut (86 types)	0.860	linear-anova-pca 0.902
pbmc (8 types)	0.913	scArches 0.931

Table S3. Per-dataset accuracy: actinn-jax against whichever method leads that dataset. † on lung_cross the exact-match score is a vocabulary artifact, so ontology concordance is the meaningful column.